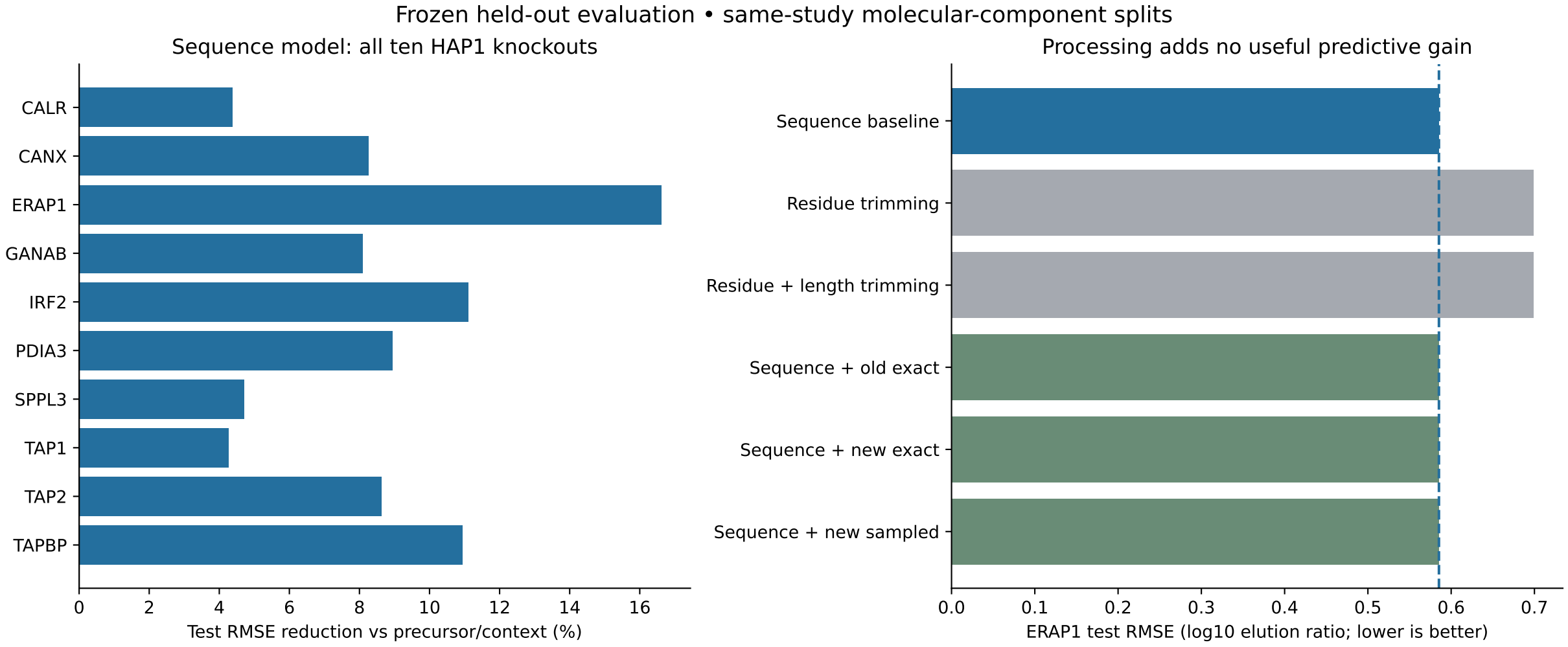
Frozen held-out evaluation • same-study molecular-component splits
Sequence model: all ten HAP1 knockouts
CALR
CANX
ERAP1
GANAB
IRF2
PDIA3
SPPL3
TAP1
TAP2
TAPBP
0
2
4
6
8
10
12
14
16
Test RMSE reduction vs precursor/context (%)
Processing adds no useful predictive gain
Sequence baseline
Residue trimming
Residue + length trimming
Sequence + old exact
Sequence + new exact
Sequence + new sampled
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
ERAP1 test RMSE (log10 elution ratio; lower is better)